THE NOVA RESEARCH CURRICULUM

Physics

Grade 8 · Term 2

Student edition

Outcome: PH-8.2 current, voltage, resistance and power; electrical safety devices.

Lesson notes

- **Current and voltage (Weeks 1–4):** current (I) is the flow of charge, measured in amperes (A) with an ammeter in series. Voltage (V) is the push from the cell, in volts (V), measured with a voltmeter in parallel. Current in series and parallel circuits.

  ▶ **Try it on NovaSchool:** Current, voltage and resistance — open **Simulations** on your dashboard.

- **Resistance and power (Weeks 5–8):** resistance (R) in ohms (Ω); V = I × R. Longer, thinner wires have more resistance. Power P = V × I in watts (W). Energy used = power × time; electricity bills in kilowatt-hours (units). Compare an LED bulb and an old filament bulb.
- **Safety (Weeks 9–12):** fuses and circuit breakers cut the current when it is too high; the earth wire protects you if a metal case becomes live. Three-pin plug wires (live, neutral, earth). Never overload sockets; keep water away; switch off at the mains before repairs. No student work on mains electricity.

Workbook pages

Page	Task
1	Draw circuits with ammeter and voltmeter.
2	Ten V = IR calculations.
3	Home appliance power and units table from a bill.
4	Safety poster: fuse, breaker, earth wire.

Question bank

#	Outcome	Question	Your answer
1	PH-8.2	Current is measured in ...	volts / ohms / amperes / watts
2	PH-8.2	An ammeter is connected in ...	parallel / series / either / the plug
3	PH-8.2	V = 12 V, R = 4 Ω. Current?	48 A / 3 A / 16 A / 8 A
4	PH-8.2	A 230 V heater draws 5 A. Its power?	46 W / 235 W / 1,150 W / 225 W
5	PH-8.2	A fuse melts when the current is too high. True or false.	True / False

#	Outcome	Question	Your answer
6	PH-8.2	The earth wire protects you when …	the bulb is off / a metal case becomes live / the cell is flat / it rains
7	PH-8.2	A longer wire has more resistance. True or false.	True / False
8	PH-8.2	A 1,000 W iron is used for 2 hours. How many units (kWh)?	0.5 / 1 / 2 / 2,000